

Application of Hidden Markov Model in Human Motion Recognition by Using Motion Capture Data

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Abstract The present study outlines an approach that describes how the statistical pattern recognition tool as Hidden Markov Model (HMM) is used to segment, extract features from different types of human common locomotion data, classify them and make a reliable probabilistic recognition. We have utilized six type of motion (climb, forward jump, jump, run, sit and walk), and each of them was composed of eight AMC data format specifically chosen from the CMU online library. The captured data were transformed from three dimensional (3D) joints trajectories into Matlab analyzable 2-Dimensional matrix using the Singular Value Decomposition (SVD) features factorization algorithm and then trained using the Baum-Welch algorithm, which has performed a classification of each motion's extracted features for computational evaluation. The maximum iterations tolerance according to the number of motion was around 0.001 for highly accurate probability. In spite of the noisiness of the hand and toe trajectories, the recognition level was pretty higher than expected. Although, this investigation doesn't take in account the different motion's trajectories speed. The overall model has shown the probabilistic efficiency of HMM when it is about segmenting and evaluating vectorized motion's features. The proposed study is suitable to any human common locomotion capture in AMC format and can be a great tool for further modeling in motion prediction.

Keywords Singular value decomposition · Hidden Markov model · Motion recognition

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1 Introduction

Through Digital human modeling usually needed for robotics intelligence and work process analysis combined with Motion Capture technologies, the scientific world is reaching a new level in understanding human functionality beyond ergonomics design. As evolution goes, being able to distinctly make the correlation between motion's specific variability and comparable set of extracted features is becoming the core concept of Motion Recognition especially when it is about segmenting unobservable parameters. Already successful in the field of speech recognition and protein modeling because of its capacity in uncertainty compensation, Hidden Markov Model (HMM) became one of the widely used stochastic approaches to capture motion's structure in order to make reliable matching templates through vector quantization for the corresponding untrained sequence. HMM has three major design basis that are evaluate the probability or likelihood of a sequence of observations given a specific model; the determination of a best sequence of the model states; and the estimation of the model parameters for an effective recognition [1]. Most popular dynamic research is vision-based recognition analysis depending on observed video sequences where each frame describes a particular body-position shape [2–4]. The advantage of those HMM built in dynamic modeling is that the motion's characteristics are extracted by frame creating analyzable symbols, however it requires a large amount of visual data assuming that all of them are noiseless. Given the high complexity of human body motion, alternative systems that do not necessitate explicit image feature detection or segmentation are welcomed.

In favor of solving these limitations, our study proposes an application where HMM is applied to Mocap AMC data format (skeletal joint trajectories displacement data) downloaded from the open CMU online library in order to perform a set of recognition. In our case, it implies to make an efficient classification of the movement type, and to reconstitute the parameterization of that movement for the evaluation sake. Through a range of frames, the algorithm is turning the motions joints trajectories into a matrix with minimal and maximal values to index the range of variation (if we consider a motion as a succession of signal coming from each joint, we can theorize that the motion's trajectories are turned into recognizable signal variation) computed by utilizing SVD (Singular Value Decomposition), that gives out a least squares estimation of the 3D joints trajectories. Although, the Baum-Welch algorithm is used to estimate the transition and emission probabilities for each Hidden Markov Model matrix sequences to make the probabilistic guess by looping through a specifically defined number of iterations considering the level of tolerance value and a maximum number of iterations defined by the number of motion sample. At the recognition level, which is divided in two sections: the single motion data where only one motion data is tested and the batch motion data where the entire group of motion is tested, the results in term of performance have shown that it is possible to recognize an executed motion based only on 3D joints

displacement data without having to take in account circumstantial appearance like background or weather.

The remainder of this paper is organized as follow: Section 2 covers the research background and related work. Afterwards, the data factorization and the experimentation process are detailed in Sect. 3. In Sect. 4, the results are presented in terms of accuracy and reliability. In Sect. 5, the overall method is discussed. Finally, conclusion and future work are drawn from the experimental results in Sect. 6.

2 Background

In the last two decades, motion recognition and scene analysis [2], have increased in the computer programming vision community, owing in part to more sophisticated use of the tools from machine learning and the availability of open large data sets. One of the pioneer researches has used HMM to recognize human action as described in [3] by distinctly classify and successfully identify the six different tennis strokes. However, in their research when the training and testing data are from different subjects, the performance of the model has the tendency to drop. HMM combined with a rotation-invariant image representation has been also used to recognize hand gestures from video sequences [5]. However, it has not taken into account the fact that the variation between instances has to be eliminated. In Kale et al.'s [6] approach, the human shape was extracted from video images and used to conceive a HMM and then make the recognition. However, the system accuracy was limited because its efficiency depends on the style of clothing the subject wears. There are abundant published studies that attempt Motion Recognition based on 3D data, especially with the recent release of the Microsoft Kinect (home base game entertainment device). The research of [7] has been inspired by the previously named approach, but instead of using a person shape, they used 3D motion capture coming from CMU library to apply HMM in the identical system framework. However, their approach was only considering one set of motion (walking). Motion recognition enables modern users to interact with computer in a natural way, even if it is still difficult and expensive to develop effective motion recognition technologies.

3 Experiment

For our study, we have utilized six types of human common locomotion, respectively: climb, forward jump, jump, run, sit and walk defined as the input of our system. Each of these batches is composed of eight amc data (48 in total) with a standard frame per second equal to 120 and most of them have been downloaded from different subjects. The skeletal positions are preprocessed using the function

amc to matrix matlab routines, where the joints trajectories (indexed in 29 IDs referring to the amc data skeleton bone assignments) are transformed into matlab matrix. The previous step is very important considering that the SVD function is a matrix-based factorization algorithm. In the first sub-section, motion specificity and two features sets corresponding to matrix motion features range standardization and matrix features factorization are extracted. Following is the outputs of HMM classifiers using Baum-Welch training algorithm and the recognition process.

3.1 Features Extraction and Factorization

The crucial task here is to identify features that carry equivalent information for data value reduction. When the amc data are turned into 2-Dimensional matrix (Table 1), they are stored in the value I that represent the initialized vector dimension size (62) according to the number of frames. The next step of our approach is the initialization of a random real matrix: $m \times n$ (Table 2), which respectively represent the number of standardized frames (transpose operator) we want to analyze according to the number of amc (column vectors), before moving to Singular Value Decomposition (SVD).

3.2 Features Factorization

SVD is closely related to the conception of diagonal symmetric matrix correlation. According to Kalman theory, finding the SVD of our six motion's matrices represented as the arbitrary matrices noted $A = \{a_1, a_2, a_3, a_4, a_5, a_6\}$, is about combining linearly the sets of extracted features to furnish the best approximation for each motion. It also implies that it exists two orthogonal matrices as U and V and a diagonal matrix denoted S such as $A = USV^T$. In that situation, U is $m \times m$ and V is $n \times n$, so that S is rectangular with the same dimensions as

Table 1 Raw data frame number with DOFs

ID	'Climb'	'Forward jump'	'Jump'	'Run'	'Sit'	'Walk'
1	4346×62	439×62	497×62	148×62	2313×62	343×62
2	4376×62	426×62	319×62	130×62	4759×62	298×62
3	1417×62	295×62	369×62	128×62	324×62	598×62
4	1201×62	407×62	400×62	137×62	303×62	494×62
5	1387×62	422×62	322×62	143×62	768×62	316×62
6	1432×62	496×62	466×62	143×62	496×62	329×62
7	1614×62	461×62	173×62	138×62	953×62	449×62
8	415×62	483×62	351×62	128×62	752×62	517×62

Table 2 Standardized raw data with DOFs

ID	'Climb'	'Forward jump'	'Jump'	'Run'	'Sit'	'Walk'
1	124×62	124×62	124×62	124×62	124×62	124×62
2	124×62	124×62	124×62	124×62	124×62	124×62
3	124×62	124×62	124×62	124×62	124×62	124×62
4	124×62	124×62	124×62	124×62	124×62	124×62
5	124×62	124×62	124×62	124×62	124×62	124×62
6	124×62	124×62	124×62	124×62	124×62	124×62
7	124×62	124×62	124×62	124×62	124×62	124×62
8	124×62	124×62	124×62	124×62	124×62	124×62

A represented in our method by a matrix of length 62×62 , which is the dimensions of each analyzed motion. The diagonal entries of S, illustrated by $S_{ii} = \sigma_i$, are manage to be nonnegative in order to decrease the magnitude. The positive ones are called singular values of A. The columns U and V left and right singular vectors for A [8]. The least square fit here is represented by the maximum and minimum values measured by SVD during the features extraction process (matrix factorization).

3.3 Classification

We aim to model human common locomotion recognition using HMM with discrete hidden states and continuous observations from each state. An HMM has the ability to learn standardize motion pattern, in the way that it round up easily to avoid zero probability and identify sequential stochastic states where the likelihood of a state predicate the previous state. During the learning process, which was an unsupervised learning, we used the Baum-Welch algorithm so that the transition with very low likelihood will not interfere in the overall sequence classification. For that purpose, we have first of all defined a standard derivation constant named *eps* equal to $1e^{-6}$ and multiplied it by the sequences 1/6 probability to find the estimation likelihood matrix of the transition guess. Since we have used different type of motion, we observed joint displacement correlation values as the one producing the hidden state because an HMM parameters are composed of:

- Transition probability a_{ij} , $1 \leq i, j \leq N$, which are the probabilities of transitioning from state q_i to state q_j in successive time slices,
- Initial probabilities π_i , $1 \leq i \leq N$, which are the probabilities of a sequence starting in any state q_i , and
- Observation probabilities $b_i(x)$, $1 \leq i \leq N$, which are the probabilities of observing x given being in state q_i .

Table 3 HMMs models database

‘Climb’	‘Forward jump’	‘Jump’	‘Run’	‘Sit’	‘Walk’
124 × 8 cell	124 × 8 cell	124 × 8 cell	124 × 8 cell	124 × 8 cell	124 × 8cell
124 × 8 cell	124 × 8 cell	124 × 8 cell	124 × 8 cell	124 × 8 cell	124 × 8 cell
124 × 8 cell	124 × 8 cell	124 × 8 cell	124 × 8 cell	124 × 8 cell	124 × 8 cell
1 × 8 cell	1 × 8 cell	1 × 8 cell	1 × 8 cell	1 × 8cell	1 × 8 cell
1 × 2 cell	1 × 2 cell	1 × 2 cell	1 × 2 cell	1 × 2 cell	1 × 2 cell

These parameters can be summarized by $\lambda = (\pi, A, B)$ where π (minmax deviation matrix) is a vector of probabilities and A (transition guess) is a matrix of probabilities. The SVD motion pattern vector matrix represents our B (Table 3) matrix.

4 Results

After carefully extracted each motion HMM model, the first recognition test applied was the motion batch test that uses the Ultra Fast Fourier Transform (UFFT) to increase signal variation compression rate and rapidly give back a recognition percentage with the name of the batch having higher predicted and recognized motion vectors. The UFFT has taken the matrix digital signal corresponding here to the minimum and maximum deviation values coming from the untrained batch and expressed it as the weighted sum of an equivalent number of frequencies. Unexpected results have been obtained using this method knowing that this process is just about giving out the percentage of the motion type with the higher predicted and recognized motion vector rate after that has been turned into a comparable signal sequence (Table 4).

The second recognition test was about evaluating each untrained amc features to the already existing ones stored in the database. At the opposite of the cluster recognition, the isolated gesture analysis performs better because the motion boundaries are clearer and it is possible for the algorithm to detect with better accuracy the points location, even if there is some noisiness. Before pointing out the motion’s appurtenance type, the modeled HMM displays the number of vectors it has recognized for the database. The average efficiency was about 66.67 % (Table 5).

Table 4 Recognition rate of jump

Motion’s name	Rate (%)
Jump	80 %

Table 5 Recognition rate experiment 2 (batch recognition test)

Number of frames						Average recognition rate
Climb	Forward jump	Jump	Run	Sit	Walk	
32	32	32	32	32	32	66.67 %

5 Discussion

We have presented an application of HMM using skeletal-based data (MoCap) to perform a set of motion recognition. The likelihood of the observed motion parameters are based on a SVD and Baum-Welch for respectively vector quantization and parameters training. Again, we choose designed 124×8 matrix structure features characteristics because we have defined the function double to expend the motion’s variation matrix so that by multiplying with SVD diagonal matrix, it gives back the best block matrix size structure. The batch-testing process was weakened by the fact that the algorithm has only recognized the jump batch motion as the higher predicted motion’s vectors batch but it has left out 20 amc over 48 at the beginning of the experimentation. The same situation happened with the single amc data test, which means that our system needs improvement. However, 28 and 32 are still more than the half of the data batch content and it could also mean that the 20 and 16 left amc’s derivation signal was too low to be capture by the signal compression method UFFT. It was also caused by the misleading of the noise coming from the hand and toe data that had decreased the rate of some motion’s matrix information. What has been made sure in that study is that the use of MoCap data facilitates the data collection and speed up the data processing part, which implies the urgency to improve the technology. On the other hand, combining SVD vector quantization to HMM has improved the efficiency of the algorithm such as from the predicted derivation features extracted vectors it had been possible to identify correctly the motion’s category. However, we realized that SVD has extracted the least square values derivation of each motion after five iterations, which was designed to be six and during the recognition process, the HMM classifier has only consider the fourth iteration of 1×8 structure as the optimal sequence for the recognition. Although, the structure 1×2 vector features variation for each type of motion was the best fitting prediction made by SVD. The obviously first relevant par of this study is the fact many HMM-based research talk about the need for a large set of training data information [9], which is true but perhaps we need more sophisticate tools.

6 Conclusion

All chosen motions sequences for the verification of our model were accessed from the CMU online library that they are all from different subjects, which primarily indicates that the model has built a strong recognizable series of pattern for each motions to be able to render out a high recognition rate. Using SVD combined with HMM has been an efficient mixture method because it took those high variables set of 3D motion data points and reduced them into a lower dimensional rate in order to exposed the substructure of the original data more clearly without loosing the motion data main correlation. Consequently, our application has revealed that motion prediction could have been also done by the fact that one of the advantageous properties of SVD is that it always deduces the optimal set of vectors that best predict the motion variation range, based on the conventional matrix similarity measurement. Even though our approach has not take in account the motions trajectories' speed, the relevant inquiry to keep in mind is to figure out how to apply these new applications to improve upcoming technological innovations.

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References

1. Rabiner, R.L.: A tutorial on hidden markov models and selected applications in speech recognition. *Proc. IEEE* **77**(2), 257–285 (1989)
2. Starner, T., Pentland, A.: Real-time American sign language recognition from video using hidden Markov models. In: *International Symposium on Computer Vision* (1995)
3. Sun, X., Chen, C., Manjunath, B.S.: Probabilistic motion parameter models for human activity recognition. In: *Proceedings of 16th international conference on pattern recognition* (2002)
4. Yamato, J., Ohya, J., Ishii, K.: Recognizing human action in time-sequential images using hidden Markov model. In: *Proceedings of IEEE computer society conference on computer vision and pattern recognition* (1992)
5. Schlensig, J., Hunter, E., Jain, R.: Vision based hand gesture interpretation using recursive estimation. In: *Proceedings of the twenty-eighth Asilomar conference on signals, systems and computers* (1994)
6. Kale, A., Cuntoor, N., Kruger, V.: Gait-based recognition of humans using continuous HMMs. In: *Proceedings of IEEE international conference on automatic face and gesture recognition* (2002)
7. Adam, A., Amershi, S.: Identifying humans by their walk and generating new motions using hidden markov models. In: *Topics in AI, The University of British Columbia* (2008)
8. Kalman, D.: A singularly valuable decomposition: the SVD of a matrix. *Coll. Math. J.* (2002)
9. Gribovskaya, E., Billard, A.: Human motion prediction in a human-robot joint task. In: *IEEE conference on intelligent robots and systems, Workshop on robotics challenges for machine learning* (2008)